THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An arc subtends an angle of 150° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 75° **(b)** 150° **(c)** 105° **(d)** 300°

2. ABCD is a cyclic quadrilateral with $\angle A = 65^\circ$. Then $\angle C$ is: 1

(a) 115° **(b)** 58° **(c)** 65° **(d)** 295°

3. The angle subtended by a diameter at any point on the circle is: 1

(a) 180° **(b)** 45° **(c)** 60° **(d)** 90°

4. Through three non-collinear points the number of circles that can be drawn is: 1

(a) two **(b)** none **(c)** exactly one **(d)** infinitely many

5. The sum of either pair of opposite angles of a cyclic quadrilateral is: 1

(a) 360° **(b)** 90° **(c)** 270° **(d)** 180°

6. The area of a parallelogram with base 15 cm and height 8 cm is: 1

(a) 60 cm^2 **(b)** 135 cm^2 **(c)** 46 cm^2 **(d)** 120 cm^2

7. The area of a triangle with base 24 cm and height 5 cm is: 1

(a) 84 cm^2 **(b)** 60 cm^2 **(c)** 120 cm^2 **(d)** 29 cm^2

8. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 7 cm is: 1

(a) 308 cm^2 **(b)** 77 cm^2 **(c)** 44 cm^2 **(d)** 154 cm^2

9. The ratio of the circumference of a circle to its diameter is: 1

(a) equal to the radius **(b)** equal to 2

(c) the same for all circles (π) **(d)** different for different circles

10. In Heron's formula, s stands for: 1

(a) the longest side **(b)** the perimeter

(c) the area of the triangle **(d)** the semi-perimeter of the triangle

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In a circle of radius 5 cm, one chord is 3 cm from the centre and another is 4 cm from the centre. Which is longer? Find both lengths. 2

12. Points C and D lie on the same side of AB and $\angle ACB = \angle ADB$. What can you say about A, B, C, D? 2

13. In $\triangle ABC$ of area 48 cm^2, AD is a median. Find the area of $\triangle ABD$ and give the reason. 2

14. Among all rectangles with perimeter 40 units, which has the greatest area? State that area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. ABCD is cyclic, $\angle BAC = 30^\circ$ and $\angle BCD = 100^\circ$. Find $\angle CBD$. 3

16. A wheel has diameter 56 cm. How many revolutions will it make to cover 1.76 km? ($\pi = \frac{22}{7}$) 3

17. A chord of a circle of radius r is at distance d from the centre. Derive the chord length. Hence find it for $r = 10$ and $d = 6$. 3

OR

In a circle of radius 10 cm there are two chords at distances 6 cm and 8 cm from the centre. Find their lengths and show that the nearer one is longer. 3

18. A parallelogram has base 20 cm, height 9 cm and one side 15 cm. Find its area and the distance between its other pair of parallel sides. 3

OR

Find the area of a trapezium with parallel sides 12 cm and 20 cm and distance 7 cm between them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the angle subtended by an arc at a point on the circle is half the angle at the centre in the case where the line DC produced meets the circle at a point E outside the arc. 5

OR

Case study: In a circle with centre O, AB is a diameter and C, D are points on the circle on opposite sides of AB. $\angle CAB = 30^\circ$ and $\angle ABD = 50^\circ$. 5

(i) Find $\angle ACB$ and $\angle ADB$.

(ii) Find $\angle ABC$.

(iii) Find $\angle BAD$.

(iv) State whether ACBD is cyclic and why.

20. **Case study:** A pizza of radius 14 cm is cut into 8 equal sectors. ($\pi = \frac{22}{7}$) 5

(i) Find the angle of each sector.

(ii) Find the arc length of one slice.

(iii) Find the perimeter of one slice.

(iv) Find the area of one slice.

OR

Case study: A cyclic plot has sides 25 m, 39 m, 52 m and 60 m. 5

(i) Find its semi-perimeter.

(ii) Find its area by Brahmagupta's formula.

(iii) Show that the plot's diagonal of length 65 m is a diameter by finding the area as two right triangles (25-60-65 and 39-52-65).

(iv) Find the cost of levelling at ₹20 per m^2.

END OF QUESTION PAPER — Total: 40 Marks